

The Schwarzschild geometry

21

As you see the war treated me kindly enough, in spite of heavy gunfire, to allow me to get away from it all and take this walk in the land of your ideas.

Karl Schwarzschild (1873–1916) *in a 1915 letter to Einstein*

After Einstein wrote down the field equation of general relativity he did not expect it to admit exact solutions owing to its complexity. He himself used an approximate solution in his 1915 article about the perihelion of Mercury. It therefore came as something of a surprise when Einstein received a letter from Karl Schwarzschild at the end of 1915 detailing a rather simple exact solution.¹ Schwarzschild was, despite being over forty years old, then serving as a soldier in World War I and carried out his work while at the Russian front.

The **Schwarzschild solution** is a solution of the Einstein field equation for the geometry *outside* a spherically symmetric, gravitating mass distribution. (We will find that the solution *inside* a mass distribution is different.) The Schwarzschild solution gives us a useful metric tensor that we will use to describe the spacetime outside stars and black holes, along with the motion of objects orbiting these bodies. First, the answer: the Schwarzschild metric line element for the space outside a static, spherically symmetric gravitating body of mass M at the origin of a set of coordinates (t, r, θ, ϕ) is written as²

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (21.1)$$

Notable features of this line element include: (i) it is static: none of the components³ $g_{\mu\nu}$ of the metric field depend on time; (ii) it is asymptotically flat: as $r \rightarrow \infty$ it looks like the Minkowski metric; and (iii) it appears badly behaved at the origin, and also when $r = 2M$, where the second term becomes singular. One additional reason why this line element is so special is that **Birkhoff's theorem**⁴ tells us that any spherically symmetric solution to the Einstein equation outside a gravitating object (they don't need to be static, for example) will be identical to Schwarzschild's static solution.

In this chapter, we will examine where the Schwarzschild solution comes from, and how we justify its form. Our results will be useful in the later chapters in this part of the book, where we will apply the

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¹In 1915, Schwarzschild started suffering from pemphigus, a rare autoimmune skin disease that likely led to his death in 1916. English speakers should consider saying 'shvarts-shilt' rather than the commonly heard 'shworts-child'. The name means black shield, rather than black child, in any case.

²Our choice of units in this part of the book is $G = c = 1$. To obtain real-world units in the metric substitute $M \rightarrow GM/c^2$ and $t \rightarrow ct$.

³The non-zero components of the metric tensor are

$$\begin{aligned} g_{tt} &= - \left(1 - \frac{2M}{r}\right), \\ g_{rr} &= \left(1 - \frac{2M}{r}\right)^{-1}, \\ g_{\theta\theta} &= r^2, \\ g_{\phi\phi} &= r^2 \sin^2 \theta. \end{aligned}$$

⁴George David Birkhoff (1884–1944). Birkhoff's theorem says that any spherically symmetric solution of the vacuum field equations must be static and asymptotically flat.

⁵We'll postpone examining the light cone structure of this metric field until then.

In non-natural units, the Schwarzschild metric is given by

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (21.2)$$

Note that, unlike the coordinate systems we met in the previous part of the book, Schwarzschild coordinates are not comoving, since $g_{tt,r} \neq 0$. Physically, this is because the Schwarzschild coordinates privilege the rest frame of the spherically symmetric mass distribution that acts as the source of the curvature.

metric and use the curvature tensor to examine motion such as orbits and objects like black holes.⁵

21.1 Justifying the solution

We seek a solution to the Einstein equation for a static, spherically symmetric mass distribution centred on the origin of a set of spherical coordinates (t, r, θ, ϕ) . Although, as we have discussed, the names of these components have no intrinsic metric significance, it will be helpful to give them some temporarily in an effort to make sensible assumptions. We start by assuming that the metric field $\mathbf{g}$ is static. This means that intervals between events are time independent and so components obey $dg_{\alpha\beta}/dt = 0$. We assume that the spherical symmetry means that world lines of constant r, θ and ϕ are orthogonal to $t = (\text{const.})$ hypersurfaces. We also assume asymptotic flatness, which is to say that as $r \rightarrow \infty$, we must have flat spacetime (that is, the gravitational forces vanish at infinity).

A good place to start is therefore with spherically symmetric, flat spacetime, which obeys the above assumptions and has a metric line element

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2, \quad (21.3)$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$. The simplest way to proceed is then to guess that these components assume different values close to the gravitating object, subject to obeying the constraints of the previous paragraph. Owing to spherical symmetry, the new components can only depend on the radius coordinate r and so we write an expression with three functions

$$ds^2 = -e^{2\Phi(r)} dt^2 + e^{2\Lambda(r)} dr^2 + R(r)^2 d\Omega^2. \quad (21.4)$$

We can immediately now restrict the number of variables from three to two. This is because we can simply reinterpret $R(r)$ as the r coordinate and rescale the other functions.⁶ So we have a general line element

$$ds^2 = -e^{2\Phi(r)} dt^2 + e^{2\Lambda(r)} dr^2 + r^2 d\Omega^2, \quad (21.5)$$

where we constrain $\Phi(\infty) = \Lambda(\infty) = 0$.

Expressions for $\Phi(r)$ and $\Lambda(r)$ will come from linking the line element in eqn 21.5 to the physics of gravitating objects, which we do below. However, we can immediately gain an insight into the function $\Phi(r)$ by comparison with the weak-field metric, which has a line element, expressed in spherical coordinates, of

$$ds^2 = -(1 + 2\Phi)dt^2 + dr^2 + r^2 d\Omega^2, \quad (21.6)$$

where Φ is the Newtonian gravitational potential. In eqn 21.5, we observe that in the limit of small Φ we have that $-e^{2\Phi} \approx -(1 + 2\Phi)$, suggesting that this is the same gravitational potential Φ that features in the weak-field metric.⁷

⁶Recall our slogan that coordinates have no intrinsic significance in the metric field, so one radius-like variable is just as good as another.

⁷This also provides us with an operational definition of when a field can be characterized as being weak (i.e. restoring factors, we want $2\Phi/c^2 \ll 1$).

Now that we have, in eqn 21.5, a candidate metric, we can feed it through the Einstein field equation $\mathbf{G} = 8\pi\mathbf{T}$. This allows us to discover that the metric does indeed solve the problem and also how to provide expressions for Φ and Λ in terms of physically meaningful quantities.

21.2 Components of the Riemann tensor

We start by finding the left-hand side of the Einstein field equation. This involves finding the curvature tensor $\mathbf{R}$ for the metric field in eqn 21.5 and then generating the Einstein tensor $\mathbf{G}$. The components of $\mathbf{R}$ can be generated in a number of ways. To maximize simplicity, we will work in the orthonormal frame.⁸

Example 21.1

Written in its general form, the spherically symmetric metric line element looks like

$$ds^2 = -e^{2\Phi} dt^2 + e^{2\Lambda} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad (21.7)$$

where Φ and Λ are functions of r only. Feeding this into the equations to find the components of the Riemann tensor, we obtain⁹

$$\begin{aligned} R^{\hat{t}\hat{r}}_{\hat{t}\hat{r}} &= E, & R^{\hat{t}\hat{\theta}}_{\hat{t}\hat{\theta}} &= \bar{E}, & R^{\hat{t}\hat{\phi}}_{\hat{t}\hat{\phi}} &= \bar{E}, \\ R^{\hat{\theta}\hat{\phi}}_{\hat{\theta}\hat{\phi}} &= F, & R^{\hat{\phi}\hat{r}}_{\hat{r}\hat{\phi}} &= -\bar{F}, & R^{\hat{r}\hat{\theta}}_{\hat{r}\hat{\theta}} &= \bar{F}, \end{aligned} \quad (21.8)$$

where

$$\begin{aligned} E &= -e^{-2\Lambda} (\Phi'' + \Phi'^2 - \Phi'\Lambda'), \\ \bar{E} &= -\frac{1}{r} e^{-2\Lambda} \Phi', \\ F &= \frac{1}{r^2} (1 - e^{-2\Lambda}), \\ \bar{F} &= \frac{1}{r} e^{-2\Lambda} \Lambda', \end{aligned} \quad (21.9)$$

and we have used the dash notation for derivatives with respect to r .

These components of the Riemann tensor $\mathbf{R}$ give us the non-zero components of the Einstein tensor¹⁰

$$\begin{aligned} G^{\hat{t}}_{\hat{t}} &= -G_{\hat{t}\hat{t}} = -(F + 2\bar{F}), \\ G^{\hat{r}}_{\hat{r}} &= G_{\hat{r}\hat{r}} = -(F + 2\bar{E}), \\ G^{\hat{\theta}}_{\hat{\theta}} &= G^{\hat{\phi}}_{\hat{\phi}} = -(E + \bar{E} + \bar{F}). \end{aligned} \quad (21.10)$$

Note that $\mathbf{G}$ will vanish outside a mass distribution, although we can have curvature (i.e. non-zero $\mathbf{R}$) this region. The tensor $\mathbf{G}$ is zero here because the Einstein equation forces $\mathbf{G}$ to be proportional to the energy-momentum tensor $\mathbf{T}$, which itself vanishes in the vacuum.

⁸Remember that one advantage of this is that components can be raised and lowered with the Minkowski tensor $\boldsymbol{\eta}$. The vielbein components for observers in the stationary orthonormal frame are

$$\begin{aligned} (\mathbf{e}_t)^{\hat{t}} &= e^{\Phi}, \\ (\mathbf{e}_r)^{\hat{r}} &= e^{\Lambda}, \\ (\mathbf{e}_\theta)^{\hat{\theta}} &= r, \\ (\mathbf{e}_\phi)^{\hat{\phi}} &= r \sin \theta. \end{aligned}$$

In the Schwarzschild case, we can safely call the observer in this frame stationary, since we can say they are stationary relative to the mass at the origin.

⁹See the exercises at the end of Chapter 11 and also Chapter 36.

¹⁰The simplest way to compute these is to use the useful rules that

$$\begin{aligned} G^0_0 &= -(R^{12}_{12} + R^{23}_{23} + R^{31}_{31}), \\ G^1_1 &= -(R^{02}_{02} + R^{03}_{03} + R^{23}_{23}), \\ G^0_1 &= R^{02}_{12} + R^{03}_{13}, \\ G^1_2 &= R^{10}_{20} + R^{13}_{23}, \end{aligned}$$

where other components can be found using cyclic permutations. You're invited to prove these rules in Exercise 21.3.

This gives us the left-hand (geometrical) side of the Einstein equation. In the next section, we look at the right-hand (physical) side.

21.3 A gravitating object

Now for the right-hand side of the Einstein equation. We make a spherically symmetric distribution of static perfect fluid so that, in the orthonormal frame, the inside of the mass distribution is described by $T^{\hat{\mu}\hat{\nu}} = \text{diag}(\rho, p, p, p)$, where ρ is the mass density and p is the pressure.¹¹ We'll take the total mass of the distribution giving rise to the metric to be M , and stipulate that the mass distribution stretches out to some maximum radius $r = R$. Outside this radius there is no matter and so the components of $\mathbf{T}$ must vanish locally.

The function $\Lambda(r)$ can be determined by considering the 00th component of the Einstein equation. We have, from the previous section, that the component $G_{\hat{t}\hat{t}} = 8\pi T_{\hat{t}\hat{t}}$ can be written as

$$\frac{1}{r^2} \frac{d}{dr} [r(1 - e^{-2\Lambda})] = 8\pi\rho. \quad (21.12)$$

We will see very shortly that it makes sense to call the quantity in the square brackets $2m(r)$, which is to say¹²

$$e^{2\Lambda} = \left(1 - \frac{2m(r)}{r}\right)^{-1}. \quad (21.13) \quad r(1 - e^{-2\Lambda}) = 2m(r). \quad (21.14)$$

Returning to the Einstein equation 21.12, this definition of $m(r)$ gives us

$$\frac{2}{r^2} \frac{dm(r)}{dr} = 8\pi\rho, \quad (21.15)$$

whose solution is

$$m(r) = \int_0^r dr \, 4\pi r^2 \rho + m(0). \quad (21.16)$$

Since ρ is a density, this expression makes physical sense for a spherically symmetrical object if we interpret $m(r)$ to be the mass contained in a sphere of radius r .

In order to find $\Phi(r)$, we need only consider the $\hat{r}\hat{r}$ component of the Einstein equation, which gives us

$$-\frac{1}{r^2} + \frac{1}{r^2} e^{-2\Lambda} + \frac{2}{r} e^{-2\Lambda} \frac{d\Phi}{dr} = G_{\hat{r}\hat{r}} = 8\pi p. \quad (21.17)$$

Substituting for Λ with $m(r)$ (from eqn 21.13), this expression becomes¹³

$$\frac{d\Phi}{dr} = \frac{m(r) + 4\pi r^3 p}{r[r - 2m(r)]}. \quad (21.19)$$

This looks complicated, but if we concentrate on the field outside of the mass distribution, where $p = 0$, then we can solve eqn 21.19 straightforwardly. In the region $r > R$, we deduce from eqn 21.16 that the mass function $m(r > R)$ must be constant. This constant must, of course, be the total mass M and so we exchange $m(r)$ for M , set $p = 0$ and write an expression valid for outside the mass distribution of

$$\frac{d\Phi}{dr} = \frac{M}{r(r - 2M)}. \quad (21.20)$$

¹¹That is, in the orthonormal frame with basis vectors $\mathbf{e}_{\hat{t}}, \mathbf{e}_{\hat{r}}, \mathbf{e}_{\hat{\theta}}, \mathbf{e}_{\hat{\phi}}$ we have

$$T^{\hat{\mu}\hat{\nu}} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix}. \quad (21.11)$$

¹²This implies that

$$e^{2\Lambda} = \left(1 - \frac{2m(r)}{r}\right)^{-1}. \quad (21.13)$$

¹³In the Newtonian limit, the potential becomes

$$\frac{d\Phi}{dr} = \frac{m}{r^2}, \quad (21.18)$$

as we might expect.

The solution to the latter expression is¹⁴

$$\Phi(r) = \frac{1}{2} \ln \left(1 - \frac{2M}{r} \right) \quad (\text{outside the mass}). \quad (21.22)$$

The resulting expressions for $e^{2\Lambda}$ and $e^{2\Phi}$ give us all we need to justify the form of the Schwarzschild line element outside a gravitating body, which is¹⁵

$$ds^2 = - \left(1 - \frac{2M}{r} \right) dt^2 + \left(1 - \frac{2M}{r} \right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (21.23)$$

Example 21.2

Substituting the more familiar variables M and r , we find

$$\begin{aligned} \Phi &= \frac{1}{2} \ln \left(1 - \frac{2M}{r} \right), & \Phi' &= \frac{M}{r(r-2M)}, & \Phi'' &= \frac{-2M(r-M)}{r^2(r-2M)^2}, \\ \Lambda &= \frac{1}{2} \ln \left(\frac{r}{r-2M} \right), & \Lambda' &= \frac{-M}{r(r-2M)}, \end{aligned} \quad (21.24)$$

and also

$$\begin{aligned} E &= \frac{2M}{r^3}, & \bar{E} &= \frac{-M}{r^3}, \\ F &= \frac{2M}{r^3}, & \bar{F} &= \frac{-M}{r^3}. \end{aligned} \quad (21.25)$$

In terms of the familiar variables M and r , the useful components of the Riemann curvature tensor are¹⁶

$$\begin{aligned} R_{\hat{t}\hat{r}\hat{t}\hat{r}} &= -\frac{2M}{r^3}, & R_{\hat{t}\hat{\theta}\hat{t}\hat{\theta}} &= \frac{M}{r^3}, & R_{\hat{t}\hat{\phi}\hat{t}\hat{\phi}} &= \frac{M}{r^3}, \\ R_{\hat{\theta}\hat{\phi}\hat{\theta}\hat{\phi}} &= \frac{2M}{r^3}, & R_{\hat{r}\hat{\theta}\hat{r}\hat{\theta}} &= -\frac{M}{r^3}, & R_{\hat{r}\hat{\phi}\hat{r}\hat{\phi}} &= -\frac{M}{r^3}. \end{aligned} \quad (21.26)$$

We also observe that components of $\mathbf{G}$ vanish as they must, since we're in a region with $\rho = p = 0$ and so $0 = 8\pi\mathbf{T} = \mathbf{G}$.

Example 21.3

The arguments in this section can also be used to derive the **Tolman-Oppenheimer-Volkov** (TOV) equations that describe the structure of a static, spherical perfect fluid, and which therefore provides us with a relativistic model of a star. Starting with the stress-energy tensor for a perfect fluid, we can use $\nabla \cdot \mathbf{T} = 0$ to show¹⁷ that in the Schwarzschild geometry

$$\frac{\partial p}{\partial r} + (p + \rho) \frac{\partial \Phi}{\partial r} = 0. \quad (21.27)$$

This equation can be combined with eqn 21.19 to give the first TOV equation

$$\frac{\partial p}{\partial r} = \frac{[p(r) + \rho(r)] [m(r) + 4\pi r^3 p(r)]}{r^2 \left[1 - \frac{2m(r)}{r} \right]}. \quad (21.28)$$

The second TOV equation gives us a quantity to be interpreted as the mass

$$m(r) = 4\pi \int_0^r dr r^2 \rho(r), \quad (21.29)$$

where we have fixed the integration constant $m(0) = 0$. The solution to these equations requires a choice of equation of state (that is, a link between p and ρ). They are generally integrated numerically from the origin outwards until $p(r = R) = 0$, which we take to give the surface of the star, with $m(R)$ giving us the stellar mass.

¹⁴This gives us

$$e^{2\Phi(r)} = \left(1 - \frac{2M}{r} \right). \quad (21.21)$$

The connection coefficients for the Schwarzschild metric are

$$\begin{aligned} \Gamma_{rt}^t &= \Phi', \\ \Gamma_{tt}^r &= \Phi' e^{2\Phi} e^{-2\Lambda}, \\ \Gamma_{rr}^r &= \Lambda', \\ \Gamma_{\theta\theta}^r &= -r e^{-2\Lambda}, \\ \Gamma_{\phi\phi}^r &= -r e^{-2\Lambda} \sin^2 \theta, \\ \Gamma_{r\theta}^\theta &= \frac{1}{r}, \\ \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta, \\ \Gamma_{r\phi}^\phi &= \frac{1}{r}, \\ \Gamma_{\theta\phi}^\phi &= \frac{\cos \theta}{\sin \theta}, \end{aligned}$$

where a dash denotes a derivative with respect to the r coordinate.

(These are computed in Exercise 9.7).

¹⁵In a letter to Schwarzschild in 1916, Einstein wrote “I had not expected that one could formulate the exact solution of the problem so simply. The analytical treatment of the problem seems to me to be excellent.”

¹⁶Achieved by lowering indices using the Minkowski tensor (i.e. lowering a $\hat{t}$ gives a minus sign, lowering everything else has no effect).

Richard C. Tolman (1881–1948).

J. Robert Oppenheimer (1904–1967).

George Volkoff (1914–2000).

Oppenheimer and Volkoff used Tolman's work as a basis that led to their prediction of the existence of neutron stars.

¹⁷See Exercise 39.5.

¹⁸As examined in the exercises, the Newtonian prediction for hydrostatic equilibrium is

$$4\pi r^2 dp = -\frac{mdm}{r^2}. \quad (21.31)$$

It will be useful in later chapters to have in mind a simple picture of **stellar evolution**. Stars are formed from gas clouds that collapse under gravity to eventually achieve equilibrium, where gravitational collapse is balanced against outward radiation pressure resulting from nuclear fusion of hydrogen in the star's core. The hydrogen in the core eventually becomes exhausted, causing many stars (i.e. those of around one solar mass $M_\odot$), to fuse helium inside their core and hydrogen outside, leading to expansion into a **red giant**. Once the nuclear fuel is exhausted, the star collapses under its own gravity, typically forming a **white dwarf**, with outer layers of mass thrown off as planetary nebula. The white dwarf achieves equilibrium with gravitation collapse now balanced by the outward pressure caused by the electronic matter making up the dwarf being **degenerate**. The latter implies the density of electronic matter is such that the quantum energy levels are completely occupied by electrons, causing them to stack up in energy owing to the Pauli principle. This leads to a large outward pressure reflecting the difficulty in rearranging particles between energy levels while preventing multiple occupancy. For stars with masses $\gtrsim 1.5M_\odot$, the white dwarf cannot achieve this equilibrium owing to the size of the inward gravitational force, and a **neutron star** or **black hole** can be formed. The former involves electrons and protons fusing to form neutrons, with equilibrium now achieved from the outward pressure of the degenerate neutron matter. Pulsars were discovered by Jocelyn Bell Burnell (1943–) in 1967, and were later identified as rapidly rotating neutron stars. Black holes are discussed in Chapter 25.

¹⁹We will often invoke the observer at infinity, whose proper time $\tau = t$.

To interpret the first equation it can be rewritten as

$$4\pi r^2 dp(r) = -\frac{m(r)dm}{r^2} \left[1 + \frac{p(r)}{\rho(r)}\right] \left[1 + \frac{4\pi r^3 p(r)}{m(r)}\right] \left[1 - \frac{2m(r)}{r}\right]^{-1}. \quad (21.30)$$

The first term on the right is the Newtonian prediction¹⁸ with the remaining terms giving the relativistic corrections. In low-mass stars, the largest contribution to ρ is from baryons (i.e. nuclear matter), which don't contribute significantly to the pressure (which turns out to be provided by electrons). This means $p/\rho \approx 0$ and $p/m \approx 0$, so that the relativistic corrections are not large. In larger stars, the pressure and energy density increase the right-hand side of the equation, steepening the pressure gradient. This ultimately causes the radius of large stars to be reduced compared to the prediction of Newtonian physics.

21.4 The meaning of the coordinates

In general relativity, coordinates have no *intrinsic* metric significance. However, we *can* relate the coordinates to the quantities of interest in describing stars and orbits in this specific case. This is our next task.

Example 21.4

Let's consider a circle in the equatorial plane ($\theta = \pi/2$) at an instant in time ($dt = 0$). We then have

$$ds^2 = r^2 d\phi^2. \quad (21.32)$$

Taking a square root and integrating, we find the circumference of the circle is

$$C = \int_{\phi=0}^{2\pi} r d\phi = 2\pi r. \quad (21.33)$$

We can therefore call r a radius, in that it supplies the correct scaling to work out the circumference of circles. We must be careful, since if we attempt to compute the distance between two events along a radial line, we obtain

$$\Delta s = \int_{r_A}^{r_B} \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} dr, \quad (21.34)$$

which will be greater than $r_B - r_A$.

We conclude that r does not give the distance from the origin in the Schwarzschild geometry. Since it does give the radius of circles it is sometimes known as the **circumferential radial coordinate**.

Next, we turn to the time. We find that the proper time between two events is

$$\Delta\tau = \left(1 - \frac{2M}{r}\right) \Delta t. \quad (21.35)$$

For $r \rightarrow \infty$ we have $\Delta\tau = \Delta t$. We conclude that the t coordinate is the time between events as measured by a clock at infinity.¹⁹

We now have a suitable metric field we can use it to examine the motion that is possible in the curved spacetime that it describes. In the next chapter, we look at general properties of motion in the Schwarzschild geometry before, in Chapter 23, turning to the question of orbits.

Chapter summary

- The Schwarzschild geometry refers to static, spherically symmetric spacetime.
- The metric field in the Schwarzschild geometry is given by the Schwarzschild line element

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (21.36)$$

- The Schwarzschild metric is static, asymptotically flat and is badly behaved at $r = 0$ and $r = 2M$.
- Birkhoff's theorem says that any spherically symmetric solution to Einstein's equation is identical to the Schwarzschild solution.

Exercises

- (21.1) Use the vielbein components to express the components of the energy-momentum tensor $\mathbf{T}$ for a perfect fluid in a (t, r, θ, ϕ) coordinate system.
- (21.2) (a) Verify eqns 21.10 using the method suggested in the text.
 (b) Using eqns 21.25, write the components of $\mathbf{G}$ in the orthonormal frame using the familiar polar coordinates and show that each component vanishes.
- (21.3) Using the symmetries of $\mathbf{R}$, prove the useful rule in Sidenote 10.
- (21.4) (a) Confirm that eqn 21.19 follows from eqn 21.17.
 (b) Show that eqn 21.19 implies eqn 21.18 in the Newtonian limit.
- (21.5) (a) By computing the relevant determinant g , compute the area of a spherical surface of fixed r and t in the Schwarzschild geometry.
 (b) Compute a circumference at fixed r , t and θ .
- (21.6) (a) Show that in Newtonian gravity, hydrostatic equilibrium requires that

$$4\pi r^2 dp = - \frac{GM dm}{r^2}, \quad (21.37)$$

where $dm = 4\pi r^2 \rho dr$.

Hint: Generalize the usual derivation of Pascal's

law of hydrostatics in a uniform gravitational field for the case of a non-uniform field.

- (21.7) *Justification of Birkhoff's theorem.* The most general spherically symmetric line element has the form

$$ds^2 = -A(r, t) dt^2 + B(r, t) dr^2 + 2C(r, t) dr dt + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (21.38)$$

- (a) Show that a transformation $t \rightarrow t + h(r, t)$, where $h(r, t)$ is some function, can be used to eliminate the cross term proportional to $dr dt$, such that we have

$$ds^2 = -e^{\Phi(r, t)} dt^2 + e^{\Lambda(r, t)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (21.39)$$

In Exercise 36.8, we compute the curvature properties of this spacetime. It follows from these that the components of the Einstein tensor are

$$\begin{aligned} G_{\hat{t}\hat{t}} &= r^{-2}(1 - e^{-2\Lambda}) + 2(\Lambda_{,r}/r)e^{-2\Lambda}, \\ G_{\hat{r}\hat{t}} &= 2(\Lambda_{,t}/r)e^{-(\Lambda+\Phi)}, \\ G_{\hat{r}\hat{r}} &= 2(\Phi_{,r}/r)e^{-2\Lambda} + r^{-2}(e^{-2\Lambda} - 1), \\ G_{\hat{\theta}\hat{\theta}} &= G_{\hat{\phi}\hat{\phi}} = \\ &= (\Phi_{,rr} + \Phi_{,r}^2 - \Phi_{,r}\Lambda_{,r} + \Phi_{,r}/r - \Lambda_{,r}/r)e^{-2\Lambda} \\ &\quad - (\Lambda_{,tt} + \Lambda_{,t}^2 - \Phi_{,t}\Lambda_{,t})e^{-2\Phi}. \end{aligned} \quad (21.40)$$

(b) Show that if we are in a vacuum, then $\Lambda(r, t)$ is independent of time.

(c) Use the remaining components of the Einstein equation to show that

$$\Lambda = -\frac{1}{2} \ln \left| 1 - \frac{2M}{r} \right|. \quad (21.41)$$

(d) Show further that

$$\Phi(r, t) = -\Lambda(r) + f(t), \quad (21.42)$$

where $f(t)$ is a function of time.

(e) Use this to prove Birkhoff's theorem, which says that the Schwarzschild geometry is the most general, asymptotically flat, spherically symmetric solution of the Einstein equation.

(21.8) Consider filling a universe described by the line element in eqn 21.5 with energy density ζ , such that we have $T_{\hat{t}\hat{t}} = 8\pi\zeta$ and $T_{\hat{i}\hat{i}} = -8\pi\zeta$.

(a) Use the results from this chapter to show that this energy density is consistent with $\Phi = -\Lambda$.

(b) Show that the line element

$$ds^2 = -(1 - H^2 r^2) dt^2 + (1 - H^2 r^2)^{-1} dr^2 + r^2 d\Omega^2, \quad (21.43)$$

is a solution to the Einstein equation. Here $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$ and H is a constant you should determine.

This solution is actually the de Sitter model (Universe 1 from Chapter 15) again, expressed in a different set of coordinates. The link is made in Exercise 18.8.